

Question Bank Part 2-BMAT201L-Linear Algebra



Fall Sem. 2025-26

Programme Name & Branch: B. Tech.

Course Name & Code: Complex Variables and Linear Algebra (BMAT201L)

Practice all the questions

Module 7

1. Find the eigenvalues and the corresponding eigenvectors of $A = \begin{pmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{pmatrix}$

Answer: eigenvalues are $\lambda = 0, 3, 15$ and the corresponding eigenvectors are

$$X_1 = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}, X_2 = \begin{pmatrix} -4 \\ -2 \\ 4 \end{pmatrix}, X_3 = \begin{pmatrix} -2 \\ 2 \\ -1 \end{pmatrix}.$$

2. For a matrix $A = \begin{pmatrix} 4 & 1 \\ 3 & 2 \end{pmatrix}$

- (i) Find the Eigen values and Eigen vectors of A
- (ii) Using Cayley-Hamilton theorem, find the inverse of A.
- (iii) Find the eigenvalues of A^T and A^{-1} .

Answer: (i) Eigen values and Eigen vectors of A are $\lambda = 1, 5$.

(ii) The characteristic equation of A is $\lambda^2 - 6\lambda + 5 = 0$.

By Cayley-Hamilton theorem, we have $A^2 - 6A + 5I = 0$.

$$\text{Therefore } A^{-1} = \frac{1}{5}(6I - A) \Rightarrow A^{-1} = \frac{1}{5} \begin{pmatrix} 2 & -1 \\ -3 & 4 \end{pmatrix}.$$

(iii) The eigenvalues of A^T are 1, 5 and the eigenvalues of A^{-1} .

3. Verify Cayley-Hamilton theorem for $A = \begin{pmatrix} 2 & 1 & 2 \\ 5 & 3 & 3 \\ -1 & 0 & -2 \end{pmatrix}$ and hence find A^{-1} if exists. Also

find the matrix B where $B = A^5 - 3A^4 - 7A^3 - A^2 + 2A + I$

Answer: Characteristic equation of A is $\lambda^3 - 3\lambda^2 - 7\lambda - 1 = 0$. Clearly,

$$A^3 - 3A^2 - 7A - I = 0 \text{ and } A^{-1} = \begin{pmatrix} -6 & 2 & -3 \\ 7 & -2 & 4 \\ 3 & -1 & 1 \end{pmatrix}. \text{ And } B = 2A + I.$$

4. Let $A = \begin{pmatrix} 3 & -2 & -1 \\ 2 & 3 & 4 \\ -2 & 0 & 5 \end{pmatrix}$. Use Cayley-Hamilton theorem to find constants a , b and c such

$$\text{that } A^4 = aA^2 + bA + cI, \text{ where } I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Answer: The characteristic equation of A is $\lambda^3 - 11\lambda^2 + 41\lambda - 75 = 0$.

By Cayley-Hamilton theorem, we have $A^3 - 11A^2 + 41A - 75I = 0$.

So,

$$\begin{aligned} A^3 &= 11A^2 - 41A + 75I \Rightarrow A^4 = 11A^3 - 41A^2 + 75A \\ &\Rightarrow A^4 = 11(11A^2 - 41A + 75I) - 41A^2 + 75A \\ &\Rightarrow A^4 = 80A^2 - 376A + 825I \end{aligned}$$

Therefore, $a = 80$, $b = -376$ and $c = 825$.

5. (a) Is the system of equations $x + 2y - z = 1$; $x - y + 3z = 6$; $2x - 3y + 2z = 3$ consistent or not?

Answer: Yes, it is consistent.

- (b) Solve the following system of equations using Gauss Jordan method.

$$x - y + 2z - 3w = -1; \quad 2x + y - 3z + w = 1; \quad 3x - 2y + z + 2w = 4; \quad x + 3y - z + 4w = 7.$$

6. Solve $x + 2y - 5z = -2$, $2x - y + 3z = 4$, $x + 3y - z = 3$ using (i) Gaussian elimination and (ii) Gauss Jordan and Gauss Jordan methods.

Answer: $x=1$, $y=1$, $z=1$

7. On a particular day, four persons went to market. First person bought one apple, two oranges, one mango and three bananas for 9 dollars, second person bought two apples, one orange, two

mangoes and one banana for 10 dollars, third person bought one apple, one orange, one mango and two bananas for 7 dollars and fourth person bought two apples, one orange, one mango and one banana for 8 dollars. Find the cost of each fruit using Gauss Jordan method.

Answer: Cost of each apple is 2 dollars, each orange is 1 dollar, each mango is 2 dollar and each banana is 1 dollar.

Module 4

1. (a) Let $R^4(R)$ be a vector space over a field R and $W = \{(x, y, z, w) \in R^4 \mid x = 2y, z = 3w - y\}$.

Verify whether W is a subspace of $R^4(R)$ or not?

- (b) Let $R^3(R)$ be a vector space over a field R and $S = \{(1, 1, 0), (1, 0, 2), (1, 1, 1)\}$ be a linear independent subset of R^3 . Is S a basis of $R^3(R)$.

Answer: (a) Yes, W is a subspace of $R^4(R)$.

- (b) For any $(x, y, z) \in R$ such that $(x, y, z) = a(1, 1, 0) + b(1, 0, 2) + c(1, 1, 1)$ which implies that $a = 2x - y - z$, $b = x - y$ and $c = z + 2y - 2x$.

Therefore, $L(S) = R^3$ and hence S is a basis of $R^3(R)$.

2. Is the set $S = \{(2, 1, 4), (1, -1, 2), (3, 1, -2)\}$ forms a basis of a vector space $R^3(R)$.

Answer: Yes, S is a basis of $R^3(R)$.

3. If W is the subspace of a vector space $R^4(R)$ generated by the vectors $(1, -2, 5, -3)$, $(2, 3, 1, -4)$, $(3, 8, -3, -5)$, then find a basis and dimension of W .

$$\text{Answer: } \begin{pmatrix} 1 & -2 & 5 & -3 \\ 2 & 3 & 1 & -4 \\ 3 & 8 & -3 & -5 \end{pmatrix} \xrightarrow[\substack{R_2 - 2R_1 \\ R_3 - 3R_1}]{R_2 - 2R_1} \begin{pmatrix} 1 & -2 & 5 & -3 \\ 0 & 7 & -9 & 2 \\ 0 & 14 & -18 & 4 \end{pmatrix} \xrightarrow{R_3 - 2R_2} \begin{pmatrix} 1 & -2 & 5 & -3 \\ 0 & 7 & -9 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Therefore, the two non-zero row vectors $(1, -2, 5, -3)$, $(0, 7, -9, 2)$ forms a basis of W and hence $\dim W = 2$.

4. Find the basis and dimension of the subspace $W = \{(x, y, z, w) \mid x+y=0, z=2w\}$ of a vector space $R^4(R)$.

Answer: The set $\{(-1, 1, 0, 0), (0, 0, 2, 1)\}$ is a basis of W and hence $\dim W = 2$.

5. Find the basis and dimension of the row space of $A = \begin{pmatrix} 1 & -2 & 5 & -3 \\ 2 & 3 & 1 & -4 \\ 3 & 8 & -3 & -5 \end{pmatrix}$.

And find the nullity (A).

Answer: The set $\{(1, -2, 5, -3), (0, 7, -9, 2)\}$ form a basis for the row space of A and hence the dimension of the row space of A is 2 (i.e., the rank of A is 2).

And nullity (A) = $4-2 = 2$.

Module 5

1. Is a mapping $T: \mathbb{R}^3(\mathbb{R}) \rightarrow \mathbb{R}^3(\mathbb{R})$ defined by $T(x, y, z) = (x+y-z, 2x-y+z, -x+y+2z)$ a linear transformation?

Answer: Yes, it is a linear transformation.

2. Which of the following functions are linear transformations from $\mathbb{R}^2(\mathbb{R})$ into $\mathbb{R}^2(\mathbb{R})$.

(i) $T(x, y) = (x^2, y)$ (ii) $T(x, y) = (x, |y|)$ (iii) $T(x, y) = (x-y, 0)$

Answer: (i) not a linear transformation (ii) not a linear transformation (iii) a linear transformation

3. Find the linear transformation $T: \mathbb{R}^2(\mathbb{R}) \rightarrow \mathbb{R}^2(\mathbb{R})$ such that $T(2, 3) = (4, 5)$ and $T(1, 0) = (0, 0)$.

Answer: $T(x, y) = \left(\frac{4y}{3}, \frac{5y}{3}\right)$.

4. Let $T: \mathbb{R}^3(\mathbb{R}) \rightarrow \mathbb{R}^3(\mathbb{R})$ be a linear transformation defined by

$T(x, y, z) = (2x, 4x-y, 2x+3y-z)$ for $x, y, z \in \mathbb{R}$. Show that T is invertible and find T^{-1} .

Answer: $T^{-1}(a, b, c) = \left(\frac{a}{2}, 2a-b, 7a-3b-c\right)$.

5. Let $T: \mathbb{R}^2(\mathbb{R}) \rightarrow \mathbb{R}^3(\mathbb{R})$ be a linear transformation defined by $T(x, y) = (x+y, x-y, y)$. Find the rank and nullity of T.

Answer: rank of T = 2 and nullity of T = 0.

6. Let $T: \mathbb{R}^3(\mathbb{R}) \rightarrow \mathbb{R}^3(\mathbb{R})$ be a linear transformation defined by

$T(x, y, z) = (y+z, x-z, -x-y)$. Find the matrix of the linear transformation T with respect to the basis $B = \{(0, 1, -1), (1, 0, -1), (1, -1, 0)\}$.

Answer: $[T]_B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$

7. Let $T: \mathbb{R}^3(\mathbb{R}) \rightarrow \mathbb{R}^2(\mathbb{R})$ be a linear transformation defined by

$T(x, y, z) = (2x + y - z, 3x - 2y + 4z)$. Find the matrix of the linear transformation T with respect to the bases $B_1 = \{(1, 1, 1), (1, 1, 0), (1, 0, 0)\}$ and $B_2 = \{(1, 3), (1, 4)\}$.

Answer: $[T: B_1, B_2] = \begin{pmatrix} 3 & 11 & 5 \\ -1 & -8 & -3 \end{pmatrix}$.

8. Let $B = \{(2, 3), (3, 5)\}$ and $B^1 = \{(4, 3), (1, 2)\}$ be the bases of a vector space $\mathbb{R}^2(\mathbb{R})$. Find the transition matrix (or, change of basis matrix) from B to B^1 .

Answer: $P = \begin{pmatrix} 11 & -1 \\ -6 & 1 \end{pmatrix}$.

Module 6

1. If $\alpha = (1, 3, -1)$, $\beta = (-1, 1, 2)$ are two vectors in an inner product space $\mathbb{R}^3(\mathbb{R})$,

find (i) $\|\alpha\|$ (ii) $\langle \alpha, \beta \rangle$ (iii) angle between α and β .

Answer: (i) $\|\alpha\| = \sqrt{\langle \alpha, \alpha \rangle} = \sqrt{11}$ (ii) $\langle \alpha, \beta \rangle = 1(-1) + 3(1) + (-1)2 = 0$

(iii) $\cos \theta = \frac{\langle \alpha, \beta \rangle}{\|\alpha\| \|\beta\|} = \frac{0}{\sqrt{11} \sqrt{6}} = 0 \Rightarrow \theta = \frac{\pi}{2}$.

2. Is $S = \{(1, -1, 1), (0, 1, 1), (2, 1, -1)\}$ an orthogonal set in an inner product space $\mathbb{R}^3(\mathbb{R})$.

Answer: Clearly every pair of distinct vector are orthogonal. Hence S is an orthogonal set in $\mathbb{R}^3(\mathbb{R})$.

3. Is $S = \left\{ \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right), \left(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}} \right) \right\}$ an orthonormal set in an inner product space $\mathbb{R}^2(\mathbb{R})$.

Answer: Clearly S is an orthogonal set and each vector in S is a unit vector.

Hence S is an orthonormal set in $\mathbb{R}^2(\mathbb{R})$.

4. Using Gram Schmidt process finds an orthogonal basis for $\mathbb{R}^3(\mathbb{R})$, given that

$\{(1, 2, 2), (-1, 0, 2), (0, 0, 1)\}$ is a basis for $\mathbb{R}^3$. Also find an orthonormal basis for $\mathbb{R}^3(\mathbb{R})$.

Answer: $\{(1, 2, 2), (-4/3, -2/3, 4/3), (2/9, -2/9, 1/9)\}$ is an orthogonal basis for $\mathbb{R}^3(\mathbb{R})$

and hence its orthonormal basis is $\left\{ \left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3} \right), \left(-\frac{2}{3}, -\frac{1}{3}, \frac{2}{3} \right), \left(\frac{2}{3}, -\frac{2}{3}, \frac{1}{3} \right) \right\}$.

5. On $P_2(R)$, consider an inner product as $\langle p(x), q(x) \rangle = \int_0^1 p(x)q(x)dx$. Apply Gram-Schmidt process to the basis $\{1, x, x^2\}$ to produce an orthogonal basis of $P_2(R)$ and hence find an orthonormal basis.